First Year B.Tech. Courses Applicable from A.Y. 2019-20

Title	**Engineering Mechanics**	Number	MEL1010
Department	Mechanical Engineering	L-T-P [C]	2-1-0 [3]
Offered for	B.Tech. 1st Year (All branches)	Type	Compulsory
Prerequisite	Nil		
Objectives			

1. To apply knowledge of statics and dynamics to particles and rigid bodies

Learning Outcomes
1. To model practical structural problems using concepts of free body diagrams and equilibrium conditions
2. To solve dynamic problems using Newton's Law, Work-Energy and Impulse-Momentum Principles

Contents
Statics [10 Lectures]: Basic dimensions in Mechanics, Law of dimensional homogeneity, Vector and Scalar Quantities, Elements of vector algebra. Moment of a force about a point/axis, Couple, Free Body Diagram, Equations of Equilibrium, Static indeterminacy, Trusses and Beams, Coulomb Friction, Surface contact friction, Transmission of power through belt. Pappus-Guldinus Theorems, second moments and product of Area, Principal axes.

Dynamics of Particles [6 Lectures]: Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion. Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles

Dynamics of Rigid bodies [12 Lectures]: Translation/Rotation of rigid bodies, Chasles' theorem, time derivative of vector for different references. Parallel axis theorems, Rotational Pure rotation of a body of revolution about its axis of revolution/combined with translation. Three dimensional rotation, moment of inertia tensor, relation between angular momentum and torque in three dimensions, Motion in non-inertial frames, centrifugal force, Coriolis force/acceleration, rate of change of vector in inertial and rotating frames.

Reference Books
1. Shames, I., (2003), *Engineering Mechanics*, Prentice Hall
2. Gross, D., Hauger, W. and Schrder J., (2012), *Engineering Mechanics*, Springer
3. Meriam, J. L., Kraige, L. G., (2002), *Engineering Mechanics*, John Wiley & Sons
4. Harbola, M K., (2013), Engineering Mechanics, CENGAGE Learning

Self Learning Material
1. Harbola, M K., *Engin*eering Mechanics, NPTEL Course Material, Department of Physics, Indian Institute of Technology Kanpur, https://nptel.ac.in/courses/115104094/
2. Lewin, W. H. G, Physics I: Classical Mechanics - Fall 1999, MIT, http://videolectures.net/mit801f99_physics_classical_mechanics/

Title	**Engineering Visualization**	Number	MEP1010
Department	Mechanical Engineering	L-T-P [C]	0-0-2 [1]
Offered for	B.Tech. 1st Year (All branches)	Type	Compulsory
Prerequisite	Nil		

Objectives
1. To inculcate how to expresses ideas of technical nature with a pragmatic intention.
2. Helping students understand the role of engineering graphics in a product design process.

Learning Outcomes
1. To distinguish between the different types of projections and indicate the dimensions.
2. To visualize, and communicate product design using graphics.

Contents

Module 1 [8 Labs]: Free hand Sketching of engineering objects, Interpretation of drawings as a visualisation and communication tool, Orthographic views, Isometric views, Sectioning of Objects, Perspective Projection, Assembly drawing, Animation of Assemblies, Computer-Aided Draughting

Module 2 [2 Labs]: Circuits and Electrical Engineering Drawings

Module 3 [2 Labs]: Introduction to Data Visualization tools, Introduction to AR/VR

Reference Books

1. Luzzader, W. J. and Duff, J. M., (2008), *Fundamentals of Engineering Drawing*, Prentice Hall
2. Bhatt, N. D., (2002), *Elementary Engineering Drawing*, Charoter Publishing
3. Bethune, J. D., (2007), *Engineering Graphics with Autocad,* Prentice Hall

Self Learning Material

1. Robi, *P.S.* , *Engineering Drawing,* NPTEL Course Material, Department of Metallurgical, Indian Institute of Technology Guwahati, https://nptel.ac.in/courses/112103019/

Title	**Engineering Realization**	Number	MEP1020
Department	Mechanical Engineering	L-T-P [C]	0-0-2 []
Offered for	B. Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. To develop awareness about the process of materials selection and manufacturing processes in product development
2. Appreciate the importance of working in multi-disciplinary team for technology development problems

Learning Outcomes

1. Basic understanding about engineering materials and manufacturing processes along with their applications.
2. Appreciation of product development process by endeavouring hands-on-activities

Contents:

*Module 1 [5 Labs] - Conventional Manufacturing***:** Demonstration of process flow in conventional manufacturing shops (4 exercises); Material selection and structure-property-performance relationships (1 exercise)

Module 2 [4 Labs] - Modern Manufacturing: Modelling and Simulation, Demonstration of process chain from design to realization of the prototype (1 exercise); Automation in Manufacturing (1 exercise); CNC Machine Tools (1 exercise); Modern Inspection techniques (1 exercise)

Module 3 [3 Labs]: Working with embedded Hardware (1 exercise), Sensor Integration (1 exercise), DC Motor Control (1 exercise)

Reference Books:

1. Degarmo, E. P., Kosher, R. A. and Black, J. T., Materials and Processes in Manufacturing, PHI
2. Kalpakjiam, S. and Schmid, S., Manufacturing Engineering and Technology, Pearson

Self Learning Material

1. http://web.mit.edu/2.810/www/lectures.html

Title	**Engineering Design I**	Number	OAE1010
Department	All Departments	L-T-P [C]	0-0-2 [1]
Offered for	B.Tech. 1st Year (All branches)	Type	Compulsory

Prerequisite	Nil		

Objectives
1. To design and analysis of engineering problems and develop the ability to visualize the solution.

Learning Outcomes
1. Approach to solve unstructured engineering design problems.

Contents
Engineering Design: Problem Definition, Background research, Identifying Performance Specifications, Ideation, Analysis of Design, Selection, Graphical Realization, Animation, Case studies on design conceptualization

Self Learning Material
1. https://www.edx.org/course/design-thinking-for-leading-and-learning-0
2. https://ocw.mit.edu/courses/engineering-systems-division/esd-051j-engineering-innovation-and-design-fall-2012/
3. https://www.coursera.org/courses?query=design%20thinking
4. https://ocw.mit.edu/courses/aeronautics-and-astronautics/16-810-engineering-design-and-rapid-prototyping-january-iap-2007/

Title	**Engineering Design II**	Number	OAE1020
Department	All Departments	L-T-P [C]	0-0-2 [1]
Offered for	B.Tech. 1st Year (All branches)	Type	Compulsory
Prerequisite	Nil		

Objectives
1. Understand to realizes conceptual design to a working prototype

Learning Outcomes
1. Develop insight to realize an engineering product

Contents
Product Realization: Selection of Materials, Modelling and Simulation, Dimensioning and Tolerance, Prototyping, Prototype Refinement, Integrating mechanical and electrical elements, controller integration and programming, Case studies on design realization

Self Learning Material
1. https://www.edx.org/course/design-thinking-for-leading-and-learning-0
2. https://ocw.mit.edu/courses/engineering-systems-division/esd-051j-engineering-innovation-and-design-fall-2012/
3. https://www.coursera.org/courses?query=design%20thinking
4. https://ocw.mit.edu/courses/aeronautics-and-astronautics/16-810-engineering-design-and-rapid-prototyping-january-iap-2007/

Title	**Mathematics I**	Number	MAL1010
Department	Mathematics	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	None		

Objectives
1. Train the undergraduate students towards basic understanding of Mathematics.

2. Provide student with sufficient knowledge in calculus, which can be used by the students in their respective fields.
3. Enable students to develop a working knowledge of central ideas of single and multivariable calculus.

Learning Outcomes

1. Impart knowledge of basic concepts like Limit, Continuity, Differentiability, Maxima and Minima, Reimann Integration and their applications.
2. Gain strong understanding of Integration in higher dimension and Vector Calculus.
3. Improve their ability to think critically, to analyze a problem and solve it using a wide array of tools.

Contents

Single variable Calculus* [17 Lectures]*:

[10 Lectures] Properties of Real number system, Sequences and series of real numbers, review of limit, Continuity and differentiability of functions, Rolle's theorem, Mean value theorems and Taylor's theorem, Maxima, minima and curve tracing.

[7 Lectures] Riemann integral, Fundamental theorem of calculus, Application to length, area, volume, surface area of revolution.

***Multivariable Calculus* [23 Lectures]:**

[7 Lectures] Functions of Several Variables, Sequences, Limit, and continuity of functions of several variables, Partial derivatives, Chain rule, Gradient, Directional Derivative and differentiability, Taylors formula for functions from R^2 to R, Tangent planes and normal, Maxima, minima, saddle points, Lagrange multipliers, Total derivative of Functions from R^n to R^m, Jacobian matrix, sufficient condition for differentiability.

[6 Lectures] Double and triple integrals, Change of variables, Vector fields, Gradient, Curl and Divergence, Curves.

[10 Lectures] Line integrals and their applications, Green's theorem and applications, Surfaces, Surface area, Surface integrals, Divergence theorem, Stokes' theorem and applications.

Text Books

1. Thomas, G. B. & Finney, R. L., (1992), *Calculus and Analytic Geometry,* 9th Edition, Addison Wesley Publishing Company
2. Ghorpade, S. R. & Limaye, B. V., (2006), *A Course in Calculus and Real Analysis,* Springer Verlag
3. Ghorpade, S. R. & Limaye, B. V., (2009), *A Course in Multivarible Calculus and Analysis,* Springer Verlag

Reference Books

1. Thomas Jr., G.B., Weir, M.D. and Hass, J.R., (2015), Thomas' Calculus, Pearson Education.
2. Apostol, T.M., (2006), Calculus: Vol-1 and Vol-2, John Wiley & Sons.
3. Ayers, F. and Mendelson, E. (2012) Schaum's Outline of Calculus, McGraw Hill.

Online Course Material

1. Dutta, J. (IIT Kanpur) Calculus of One Real Variable, https://nptel.ac.in/courses/109104124/2
Gupta, S.K. (IIT Roorkee) Multivariable Calculus, https://nptel.ac.in/courses/111107108/

Title	**Mathematics II**	Number	MAL1020
Department	Mathematics	L-T-P [C]	3-1-0 [4]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	None		

Objectives

The Instructor will:

1. Provide understanding of fundamentals of Linear Algebra.
2. Provide understanding of fundamentals of Differential Equations.

Learning Outcomes

The students are expected to have the ability to:

1. Impart knowledge of different structures and their properties like, Dependence, Basis and Dimension.

2. Gain strong understanding of linear transformations between structures and its relation with matrices.
3. Solving analytical and series solution for ordinary differential equations and analyzing the stability of linear ODE systems.

Contents

***Linear Algebra* (20 Lectures):**

[10 Lectures] Matrices, Vector spaces (over the field of real and complex numbers), subspaces, linear dependence/ independence, basis, dimensions, coordinate with respect to a basis, complementary subspaces, Linear transformations, Range space and rank, Null space and nullity, matrix representation of linear transformation, change of basis and similarity, rank-nullity theorem.

[10 Lectures] Inner product, Norm, Gram-Schmidt orthogonlization process, orthonormal bases. Eigenvalues and eigenvectors, characteristic polynomials, Cayley-Hamilton theorem, properties of eigenvalues and eigenvectors, diagonalization of matrices.

***Differential Equations* (20 Lectures):**

[8 Lectures] First Order Ordinary Differential Equations: Geometrical interpretation of solution, Solution methods for separable equations, Exact equations, Linear equations, Existence and uniqueness of solution of IVP, Wronskian and general solution of nonhomogeneous equations, Picards Theorem for IVP (without proof). Second Order Linear differential equations: General solution of homogeneous equation, Euler-Cauchy Equation, Extensions of the results to higher order linear differential equations.

[12 Lectures] Power Series Method, Frobenius Method, S.L.P and orthogonal functions, System of first order linear ODE and its stability.

Textbook

1. Axler, S., (1997), Linear Algebra Done Right (Undergraduate Texts in Mathematics), Springer.
2. Kreyszig, E., (2010), Advanced Engineering Mathematics, John Wiley & Sons
3. Simmons, G.F., (2002), Differential Equations with applications and Historical Notes, Tata McGraw Hill.
4. Boyce, W.E. and DiPrima, R.C., (2006) Elementary Differential Equations and Boundary Value Problems, John Wiley & Sons.

Reference Books

1. Strang, G., (2006), Linear Algebra and its Applications, Thomson, Brooks/Cole.
2. Kumaresan, S., (2000) Linear Algebra - A Geometric Approach, PHI Learning.
3. Coddington, E.A., (1961), An introduction to ordinary differential equations, Prentice-Hall.
4. Lipschutz, S. and Lipson, M. (2012) Schaum's Outline of Linear Algebra, McGraw Hill.
5. Bronson, R. and Costa, G. (2010) Schaum's Outline of Differential Equations, McGraw Hill.

Online course Material

1. Rana, I.K. (IIT Bombay), Basic Linear Algebra, https://nptel.ac.in/courses/111101115/
   Manam, S. (IIT Madras), Differential Equations for Engineers, https://nptel.ac.in/courses/111106100/1

Title	**Electromagnetism and Optics**	Number	PHL1010
Department	**Electromagnetism and Optics**	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. Provide an understanding of electric and magnetic field distribution and electromagnetic nature of radiation
2. Provide an understanding of wave propagation in different media

Learning Outcomes

1. Relate theoretical concepts with problem solving approach in electromagnetism and optics
2. Solve field distribution in vacuum, conducting and dielectric systems

Contents

ELECTROMAGNETISM:

*Electrostatics [10 Lectures]:*Calculus of Fields, Coulomb's law, Gauss's theorem, electrostatic potential, Laplace's equation, conductors, capacitors and dielectrics, boundary conditions

Magnetostatics[7 Lectures]: Biot-Savart's law, Ampère circuital law, Lorentz force, magnetism

Magnetic Induction [5 Lectures]: Faraday's law, Lenz's law, self and mutual inductance, energy stored in magnetic field, boundary conditions

Electrodynamics[8 Lectures]: displacement current, electromagnetic waves, plane wave solutions of Maxwell's equations, Poynting vector, Maxwell's equations in conductors and dielectrics, introduction to waveguides

OPTICS:

*Wave Nature of Light [12 Lectures]:*interference; Fresnel and Fraunhoffer diffraction; ordinary and extraordinary rays; plane, circular and elliptically polarized light; birefringence; half wave plates; introduction to optical microscopy: polarization, phase contrast and confocal microscopy

Textbook

1. Ghatak, A.K., (2017), *Optics*, 6th Edition, McGraw Hill Education India
2. Griffiths, D.J., (2015), *Introduction to Electrodynamics*, 4th Edition, Pearson Education India

Reference Books

1. Ida, N., (2015), Engineering Electromagnetics, 3rd Edition, Springer Nature
2. Hecht, E., (2017), *Optics (Schaum's Outline Series)*, McGraw Hill Education India
3. Born, M. and Wolf, E., (1999), Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light, 7th Edition, Cambridge University Press.

Self Learning Material

1. Ghosh, D.K., *Electromagnetic Theory*, Department of Physics, Indian Institute of Technology Bombay, https://nptel.ac.in/courses/115101005/

2. *Electricity and Magnetism*, Department of Physics, Massachusetts Institute of Technology, MIT OpenCourseWare, Course Number 8.02, (Spring 2007).: https://ocw.mit.edu/courses/physics/8-02-physics-ii-electricity-and-magnetism-spring-2007/#, License: Creative Commons BY-NC-SA.

Title	**Physics Laboratory**	Number	PHP1010
Department	Physics	L-T-P [C]	0-0-2 [1]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. Provide understanding on usage of electrical, mechanical and optical systems for various measurements.
2. Provide understanding on applying analytical techniques, error estimation and graphical analysis to the experimental data.

Learning Outcomes

1. Understand interplay of electricity, magnetism and light
2. Develop idea of linear, rotational and vibrational motion and mechanics

Contents

Laboratory Experiments

1. e/m ratio: Determination of charge-to-mass ratio of a charged particle
2. Basic current balance: Understanding electromagnetic forces
3. Dispersive power: Measuring the dispersive power of a prism using spectrometer
4. Diffraction of light wave: Examining the diffraction and interference pattern using single slit and double slits
5. Fresnel bi-prism: Determination of wavelength of monochromatic light source
6. Newton's rings: Study of Newton's rings in thin air-film
7. Hall effect: Determination of charge carrier density
8. Faraday rotation: Studying the effect of a magnetic field on the transmission of a linearly polarized light
9. Air Track and Friction Experiments
10. Moment of Inertia of Flywheel
11. Vibration of string

Text Books

1. Electromagnetism and Optics Laboratory Manual, Indian Institute of Technology Jodhpur.

Reference Books

1. Shukla, R.K. and Srivastava, A., (2006), *Practical Physics*, New Age International

Title	**Chemistry**	Number	CYL1010
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. Relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving
2. Provide knowledge of engineering materials metals, composites, polymers, cement, steel, fuels, batteries and drugs.

Learning Outcomes

1. Understanding the behavior of matter and materials using fundamental knowledge of their nature.
2. Understanding the basic concepts and implementation of the same in advanced and interdisciplinary areas.
3. Predict potential complications from combining various chemicals or metals in an engineering setting.

Contents

Atomic and Molecular Structure [6 Lectures]: particle in a box, angular momentum, wavefunction, multielectron atoms, homonuclear and heteronuclear diatomic molecules, non-covalent interactions

Solid State Chemistry [6 Lectures]: Lattice and Unit Cells, X-Ray Diffraction, Bands in solids, semiconductors, Introduction to nanomaterials based on dimensionality and properties.

Surface Chemistry[3 Lectures]: Surfaces and interfaces, surface modification, surface energy, contact angle and surface tension.

Electrochemical Systems [5 Lectures]: Electrochemical cells and EMF, Applications of EMF measurements: Steady state approximation, Chain reactions, photochemical kinetics, Fuel Cells and Batteries, OER and HER reactions, Corrosion

Basic Spectroscopy [7 Lectures]: Selection Rules, Fundamentals FTIR, Raman , UV-visible and Fluorescence spectroscopy

Transition Metal Chemistry [5 Lectures]: Colors in Complexes, Organometallics and Catalysis, Bioinorganic Chemistry

Organic Chemistry [8 Lectures]: Name Reactions, Stereochemistry and their Applications, Acid-Base Chemistry, Use of Buffers, Organic Chemistry examples such as polymers and biopolymers, drugs and photoactive molecules in everyday life and advanced technologies.

Textbooks

1. Silberberg, M., *Chemistry: The Molecular Nature of Matter and Change*, 6th Edition, McGraw Hill Education.
2. West, A.R., (2015), *Solid State Chemistry and Its Applications*, 2nd edition, John Wiley & Sons.
3. J. D. Lee, Concise Inorganic Chemistry, 5th Edition.
4. Atkins, Physical Chemistry, 11th Edition.

Reference Books

1. McMurry, J. E. & Fay, R. C. *Chemistry*, 5th Edition, Pearson.
2. Jonathan Clayden, Nick Greeves, Stuart Warren, Organic Chemistry, 2nd Edition.
3. F. Albert Cotton, Geoffrey Wilkinson, Basic Inorganic Chemistry, Wiley Student Edition.

Self Learning Material

1. https://nptel.ac.in/downloads/122101001/

Title	**Chemistry Laboratory**	Number	CYP1010
Department	Chemistry	L-T-P [C]	0-0-2 [1]
Offered for	B. Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. relate the fundamental principles of chemistry with practical problems encountered for engineers and place the emphasis on problem-solving
2. provide knowledge of engineering materials such as metal complexes, composites,dyes, polymers, surfactants steel, batteries and drugs.

Learning Outcomes

Students will be able to

1. develop synthetic and analytical laboratory skills and think innovatively.
2. understand and analyse physical concepts, interpret experimental results and relate to the applications.

Laboratory Experiments

1. Preparation of complex salt of $(Co(en)_6)Cl_3$
2. Preparation of Nylon-6,6
3. Fourier transform infrared (FTIR) spectroscopy in combination technique for analysis of caffeine in tea & coffee
4. The excitation and emission spectrum of Curcumin in solvents ethanol and hexane respectively and find stokes shift using Fluorescence Spectroscopy
5. Surface contact angle, surface tension and surface free energies of different solvents and substrates respectively.
6. Study of a reversible redox reaction using Cyclic Voltammetric analysis.
7. Study of Color in Complexes using UV visible Spectroscopy
8. Determination of Critical Micelle Concentration of a Surfactant by Conductivity Method
9. Study of Corrosion of Stainless Steel using Tafel Analysis.
10. Determination of the Enantiomeric Purity of Naproxen and Ibuprofen.

Textbooks

1. Practical Physical Chemistry (Latest ed.), By B.D. Khosla, A. Gulati & V.C. Garg
2. Laboratory Manual on Engineering Chemistry (Latest ed.), By S.K. Bhasin and Sudha Rani

Reference Books

1. Hill, R. H. & Finster, D. (2010) *Laboratory Safety for Chemistry Students,* Wiley.
2. Douglas A Skoog, F. James Holler and Stanley R. Crouch, Principles of instrumental Analysis, 6 th Edition, Cengage.

Self Learning Material

1. https://nptel.ac.in/courses/122101001/37
2. http://vlab.co.in/broad-area-chemical-sciences

Title	**Introduction to Electrical Engineering**	Number	EEL1010
Department	Department of Electrical Engineering	L-T-P [C]	3-0-2 [4]
Offered for	B.Tech (All)	Type	Compulsory
Prerequisites	--		

Objectives

1. To train the undergraduates towards basic concepts of electrical circuits, dc and ac machines.
2. To introduce the basics of solid-state electronics and their applications to undergraduate students of all disciplines

Learning Outcomes

1. Ability to analyze circuits, some electric machines, ac supply systems, and magnetic circuits.
2. Ability to analyze and design basic electronic circuits using passive and active components for various applications

Contents

Basic Circuit Concepts [4 *Lectures*]: Review of circuit laws, Thevenin's and Norton's Theorem; Star-Delta conversion, Time response of R-L-C circuits

Alternating Electrical Quantities [3 *Lectures*]*:* Concepts of phasors, impedance, real and reactive power, impedance and power diagrams

Magnetic Circuits [2 *Lectures*]: Basics of magnetic circuits, series and parallel magnetic circuits

Single Phase Transformer [3 Lectures]: Working principle, equivalent circuit, efficiency and regulation, SC and OC tests
Electro-Mechanical Energy Conversion and DC Machines [5 *Lectures*]: Principles of energy conversion, types of dc generators and motors and their characteristics, speed control of DC motors
Diodes [3 *Lectures]:* Principle of operation, diode characteristics, diode circuits and applications
Transistors [5 *Lectures*]: Principle of operation, concept of biasing, characteristics and applications of transistor, basic amplifier configurations
Operational Amplifiers [5 *Lectures*]: Basic model and op-amp specifications, closed-loop negative feedback inverting and non-inverting amplifiers, concept of virtual ground, op-amp applications.
Digital Logic [9 *Lectures*]: Logic gates, boolean algebra, canonical forms of boolean function representation, minimization of boolean functions using K-maps, combinational and sequential circuit design
Transducers [3 *Lectures*]*:* Inertial sensors*,* capacitive, magnetic, hall-effect sensors, ultrasonic transducers, temperature, bridge circuits

Laboratory

- Verification of Thevenin's Theorem in AC circuits
- Transformer assembly and equivalent parameters (using OC/SC test)
- Speed control of a DC Motor
- Diode rectifiers and analysis of ripple factors with and without filters
- BJT application as a switch (e.g. automatic street light) or amplifier (eg. audio amplifier)
- Opamp applications (e.g. Schmitt trigger as a comparator; operation of instrumentation amplifier etc.)
- Digital logic and design (Logic implementation with gates/multiplexer, counter design)
- Introduction to Microcontrollers (Arduino/Raspberry Pi) with basic experiments on controlling LED brightness/sensor integration, etc.

Textbooks

1. Rizzoni, G., (2010), *Principles and applications of Electrical Engineering*, 5th Edition, McGraw Hill Education
2. Boylestad, R. L. & Nashelsky, L., (2009), *Electronic Devices and Circuit Theory*, 10th Edition, Prentice Hall
3. Sedra, A. S. & Smith, K. C., (2011), *Microelectronic Circuits*, 6th Edition, Oxford University Press
4. Smith, R. J, and Dorf, R. C. (2009), *Circuits, Devices and Systems*, 5th Edition, Wiley
5. M. M. Mano, (2001), *Digital Design*, 3rd Edition, Prentice Hall

Reference Books

1. Kothari, D. P. and Nagrath, I. J., (2008), *Basic Electrical Engineering*, 3rd Edition, Tata McGraw-Hill, New Delhi
2. Fitzgerald, A.E., Kingsley (Jr.), C., and Umans, S.D., (2003), *Electric Machinery*, 6th Edition, Tata McGraw-Hill (McGrawHill Series in Electrical Engineering)
3. Hayt, W. H., Kemmerly, J. E., and Durbin, S. M., (2010), *Engineering Circuit Analysis*, 7th Edition, Tata McGraw Hill

Self Learning Material

Umanand, L., Electrical Technology, NPTEL Course Material, Department of Electrical Engineering, Indian Institute of Science, Bangalore, http://nptel.ac.in/courses/108108076/

Title	**Introduction to Computer Science**	Number	CSL1XX0
Department	Computer Science and Engineering	L-T-P [C]	3-0-2 [4]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. Provide an understanding of the role of computation in problem solving.
2. Provide an overview of the computing and networking environment where code gets deployed.

Learning Outcomes

1. Use the basic elements of computer programs, such as, statements, variables, events, conditionals, loops, arrays, and write programs.
2. Solve problems with software tools.

Contents

Introduction [5 Lectures]: Basics of computer organization: illustration using Arduino, Operating System, Computational Thinking: Problem solving, system design, algorithms.

Basics of Data Representation [7 Lectures]: Binary representation: Integer, Fixed point and floating point numbers, variables, arrays and linked-list, data-types

Programming Constructs [5 Lectures]: Imperative programming, statements, control constructs, procedures, Examples using Python

Object oriented Programming [10 Lectures]: Concept of class, instances, encapsulation, *inheritance*, static Methods and class Methods, object memory management, illustrations using Python. Lists, Tuples, Sets, Dictionaries, Generators Modules, packages, containers.

Programming Paradigms [15 Lectures]: Imperative programming, Declarative programming, The lambda operator, Procedural programming, Dynamic typing, Polymorphism Dynamic Binding and Duck Typing, Illustrations using Python.

Laboratory Experiments
Programming exercises using Python.

Textbook
1. Brown,M.C. (2018), *Python: The Complete Reference*, 4th Edition, McGraw Hill Education
2. Patt, Y. (2017), *Introduction to Computing Systems*, 2nd Edition, McGraw Hill Education
3. Guttag, J. (2016) Introduction to Computation and Programming Using Python: With Application to Understanding Data. MIT Press.

Reference Books
1. Mermin,N.D. (2007), *Quantum Computer Science: An introduction*, Cambridge University Press
2. Satemos, C., (2014), *Client-Server Web Apps with Javascript and Java*, 1st Edition, O'Reilly

Self Learning Material
1. *John Guttag.* 6.00SC Introduction to Computer Science and Programming. *Spring 2011. Massachusetts Institute of Technology*: MIT *OpenCourseWare, https://ocw.mit.edu. License: Creative Commons BY-NC-SA.*

Title	**Introduction to Bioengineering**	Number	BBL1XX0
Department	Bioscience & Bioengineering	L-T-P [C]	3-0-2 [4]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives
1. Provide a strong foundation of biology in the context of engineering by highlighting key differences in the approaches of both disciplines, while illustrating how bioengineering utilizes the best of both disciplines.
2. provide multidisciplinary learning

Learning Outcomes
The students are expected to have the ability to:
1. Approach biology from an engineering perspective by gaining an understanding of how biological systems function
2. Be inspired from biological systems and processes that have been perfected by evolution 3. Conceptualize novel engineering solutions for complex biological problems

Contents
Biology & Engineering Approaches [3 lectures]: Observations; deductive, inductive and abductive reasoning; hypothesis testing; reductionism; systems biology; modeling; tinkering with biological systems; cause & effect; translational research; attributes of a bioengineer
Origin & Evolution of Life [7 lectures]: Chemical basis of life, water sustains life, biomolecules, biological macromolecules, unity in diversity among life forms, phylogeny
Functioning of Biological Systems [8 lectures]: From organisms to cells, structure-function relationships, free energy, entropy, reactions, biocatalysis, energy coupling, effects of process failure
Probability in life processes [7 lectures]: Storage of genetic information, central dogma of biology, cell division, inheritance, genetic variation
Control in living systems [6 lectures]: Gene expression logic gates, control of cell division, hormones, signals & receptors, feedback control, control failure & disease
Responses of Biological systems [4 lectures]: Interactions, communication, stress responses, tolerance, adaptation, rhythms & cycles, ecosystems

Exploitation of living systems [5 lectures]: Antibiotics, Recombinant DNA Technology , Biomaterials-bioplastics and materials for healthcare, Biodiesel and drop in biofuels from algae.

Laboratory Experiments

Laboratory safety and introduction to general laboratory practices, measuring liquids in small volumes, solutions & buffer preparation, quantifying simple biomolecules, stress response, selection, biocatalysis, separation of biomolecules, microscopy.

Textbook

1. Urry L.A., Cain M.L., Wasserman S.A., Minorsky P.V., Reece J.B., (2016), *Campbell biology*. 11th Edition, Pearson.

Reference Books

1. Glick B.J., Pasternak J.J., Patten C.L., (2010), *Molecular Biotechnology: Principles and Applications of Recombinant DNA*, 4th Edition, American Society for Microbiology.
2. Johnson A.T., (2010), *Biology for Engineers*, CRC Press

Self Learning Material

1. Suraishkumar G.K., & Madhulika Dixit, *Biology for Engineers and other non-Biologists,* NPTEL Course Material, Department of Biotechnology, Indian Institute of Technology Madras, http://nptel.ac.in/courses/121106008/
2. Vishal Trivedi, *Basic Biology*, Department of Biotechnology, Indian Institute of Technology, Guwahati, http://nptel.ac.in/courses/122103039/
3. Vishal Trivedi, *Fundamentals of Biotechnology*, Department of Biotechnology, Indian Institute of Technology, Guwahati, http://nptel.ac.in/courses/102103045/

Title	**Communication Skill I**	Number	HSN1010
Department	Humanities and Social Sciences	L-T-P [C]	0-0-2 [2]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. Facilitate students to gain language competence in order to communicate effectively in a spectrum of contexts.
2. Focus on the two basic skills of language learning, namely: Listening and Reading
3. Train the student to employ the above skills in both personal and professional settings

Learning Outcomes

1. Distinguish between hearing and listening, and to understand *listening* as a skill of critical significance.
2. Engage effectively in classroom situations and also with various texts, with useful reading strategies
3. Integrate classroom learning into an everyday communicative activity, thus making the learning both reflective and experiential.

Contents

Listening: Barriers to Listening, Active Listening, Listening and Speaking, Listening and Receiving, Listening and Understanding, Listening and Responding, Listening and Remembering, Listening and Paraphrasing, Listening and Reflecting, Listening to Pre-recorded audiovisual aids, Listening to and comprehending audio-visual media including films, radio programs, academic and/or public speeches.

Reading: Introduction and application of reading techniques, Comprehension, Types of reading including Scanning, Skimming, Critical Reading and Speed Reading, exploration of academic/journalistic/creative texts through various reading strategies.

Laboratory Sessions: Will include individual and group activities and hands on experiential learning modules

1) Workshop Session on Listening with Janaki Sabesh (Storyteller, Theatre Resource Person, Voice Artiste from *Golpo Tales Unlimited*) - 4 hours
2) Workshop on Reading, Review and Critical Analysis
   (i) Film Screening and Discussion - 4 hours
   (ii) Reading Books, Newspaper Articles, Journals - 4 hours

Textbook

All reading material for the course will enhance both creative and critical thinking and will include the repertoire-newspaper articles, famous speeches, biographies, passages /narratives from science and technology, fiction, nonfiction and film, word games and puzzles and will be compiled by the instructors and shared with the student.

Reference Books

1. Bolton. D. (2008). *English Grammar in Steps: English Grammar presented, explained and practiced in context.*Orient Longman. New Delhi: India
2. Wren PC, Martin H. (2018) *High School English Grammar & Composition*. *Revised by N.D.V. Prasad Rao.* S Chand Publishing Blackie. ELT Books, New Delhi: India.
3. Wren PC, Martin H.(2018) *Key to Regular &Multicolor Edition of High School English Grammar &Composition. Revised by N.D.V. Prasad Rao*.S.Chand Publishing Blackie.ELT Books, New Delhi: India.
4. Hewings, Martin (2008) *Advanced Grammar in Use.* Cambridge University Press.

Self Learning Material

1. Iqbal, Ayesha.*English Language for Competitive Examinations*.NPTEL Course Material, Department of Humanities and Social Sciences, Indian Institute of Technology-Madras, https://nptel.ac.in/courses/109106116/
2. Choudhary, Shreesh. *Better Spoken English.* NPTEL Course Material, Department of Humanities and Social Sciences, Indian Institute of Technology-Madras, https://nptel.ac.in/courses/109106067/.

Title	**Communication Skill II**	Number	HSN1020
Department	Humanities and Social Sciences	L-T-P [C]	0-0-2 [2]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

1. Facilitate students to gain language competence in order to communicate effectively in a spectrum of contexts.
2. Focus on the two basic skills of language learning, namely: Speaking and Writing.
3. Train the student to employ the above skills in both personal and professional settings

Learning Outcomes

1. Speak more efficiently and explore the act of writing as an active process.
2. Articulate with clarity and use better language, well-equipped for specific situations.
3. Integrate classroom learning into an everyday communicative activity, thus making the learning both reflective and experiential.

Contents:

Speaking: Modes of Communication - Formal and Informal, Verbal and Non-verbal Communication, Group Discussions, Mock Personal Interviews, Powerpoint presentations (academic and professional/business), Body Language, Basic Phonetics/Pronunciation, and Telephone Etiquette.

Writing: Paragraph Writing, Writing an Email, Note Making, Report Writing, Letter Writing, Reviewing articles, Structuring academic/professional writing

Grammar: Identifying Common Errors in English, Use of Articles, Use of Prepositions, Tenses, Spellings, Filling in Worksheets.

Laboratory Sessions: Will include individual and group activities and hands on experiential learning modules

1) Practice Workshop on Speaking (Group Discussion, Personal Interview) - 4 hours
2) Writing Workshop (Essay, Creative Nonfiction/Advertising, Storyboard, Journal Writing, Letter & Email Writing) - 4 hours
3) Field Visit and Workshop (Photo Essay, Report Writing) - 4 hours

Textbook

All reading material for the course will enhance both creative and critical thinking and will include the repertoire-newspaper articles, famous speeches, biographies, passages /narratives from science and technology, fiction, nonfiction and film, word games and puzzles and will be compiled by the instructors and shared with the student.

Reference Books

1. Bolton. D. (2008). *English Grammar in Steps: English Grammar presented, explained and practiced in context.* Orient Longman. New Delhi: India

2. Wren PC, Martin H. (2018) *High School English Grammar & Composition*. *Revised by N.D.V. Prasad Rao.* S Chand Publishing Blackie. ELT Books, New Delhi: India.
3. Wren PC, Martin H.(2018) *Key to Regular &Multicolor Edition of High School English Grammar & Composition. Revised by N.D.V. Prasad Rao*.S. Chand Publishing Blackie. ELT Books, New Delhi: India.
4. Hewings, Martin (2008) *Advanced Grammar in Use.* Cambridge University Press.

Self Learning Material

1. Iqbal, Ayesha. *English Business English Communication*. NPTEL Course Material, Department of Humanities and Social Sciences, Indian Institute of Technology-Madras, https://nptel.ac.in/noc/individual_course.php?id=noc18-hs18
2. Ravichandran T. Communication Skills. NPTEL Course Material, Department of Humanities and Social Sciences, Indian Institute of Technology- Kanpur, https://nptel.ac.in/courses/109104031/

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Title	**Social Connect & Responsibilities - I** **Social Connect & Responsibilities - II**	Number	OSN1010 OSN1020
Department	Office of Students	L-T-P [C]	0-0-1 [1]
Offered for	B.Tech. 1st Year	Type	Compulsory
Prerequisite	Nil		

Objectives

The course will:
1. Provide a formal platform to students to communicate and connect to the surrounding
2. Enable to create a responsible connect with the society

Learning Outcomes

The students are expected to have the ability to:
1. Understand social responsibility
2. Practice sustainability and creativity
3. Showcase planning and organizational skills

Contents

The course is mainly activity based that will offer a set of activities for the student that enable them to connect with fellow human being, nature, Society and world at large. The course will engage students for interactive sessions, open mic, reading group, storytelling session and semester long activities conducted by faculty mentors. In the following a set of activities planned for the course have been listed:

Plantation and adoption of a tree: plantation of a tree that will be adopted for four years by a group of B.tech students. They will also make an excerpt either as a documentary or a photo blog describing the plant's origin, its usage in daily life, its appearance in folklore and literature.

Heritage walk and crafts corner: Heritage tour, knowing the history and culture of the city, connecting to people around through their history, knowing the city and its craftsman, photo blog and documentary on evolution and practice of various craft forms.

Organic farming and waste management: usefulness of organic farming, wet waste management in neighboring villages and implementation in the campus

Water conservation: Knowing the present practices in the surrounding villages and implementation in the campus, documentary or photo blog presenting the current practices

Food walk: City's culinary practices, food lore and indigenous materials of the region used in cooking.

Jamming session, open mic and poetry: Platform to connect to others. Share life stories with others

Activities

A total of 14 hrs engagement per semester is required for the first year of the B.Tech program. The students will be divided into 10 groups of 35 each. Each group will be handled by two faculty mentors. Faculty mentors will design the activities.